SankhyaVox

Saṇkhyā Voṣ — Numbers Through Voice

saṇkhyā = number · vox = voice

Minimal-Vocabulary Sanskrit Spoken Digit Recognition Using HMM-Based Grammar-Constrained Sequence Decoding

Project Type: Advanced Speech Processing Research Project

Core Challenge: Recognise Sanskrit numerals 0–99 from only 13 base words

Key Innovation: Grammar-constrained HMM exploiting Sanskrit's compositionality

Target Accuracy: 90%+ on 0–99 with mobile-recorded speech

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Abstract

SankhyaVox is a constrained automatic speech recognition (ASR) system designed to recognise spoken Sanskrit numerals in the range 0–99 from a deliberately minimal recorded vocabulary of only **13 base words**. The core thesis is that Sanskrit's inherent compositional morphology, in which larger numbers are formed from smaller atomic units, can be exploited to build a grammar-constrained Hidden Markov Model (HMM) pipeline that *generalises far beyond its training vocabulary* without requiring additional data collection.

Data are collected from **10 or more speakers** using mobile phones in quiet rooms, using a structured repeated-utterance protocol that enables automated audio segmentation via energy-based silence detection. Acoustic features consist of 39-dimensional MFCC vectors ($13 + \Delta + \Delta\Delta$). Word-level left-to-right HMMs with Bakis topology are trained using Baum-Welch (EM); inference uses Viterbi decoding constrained by a formal context-free grammar over the digit vocabulary.

Baseline comparisons against GMM, k-NN, and SVM classifiers demonstrate the advantage of temporal sequence modelling. Under clean, controlled conditions, we target **90–96% word accuracy** on 0–99 with new unseen speakers. The project contributes a novel mobile-recorded Sanskrit digit dataset, a reusable automated segmentation pipeline, and a reproducible grammar-constrained HMM baseline for low-resource ancient-language ASR.

Key Novelty at a Glance

- **Compositionality exploit:** 13 words → full 0–99 coverage via grammar
- **Mobile-native corpus:** No studio, no proprietary microphone — reproducible by any student
- **Auto-segmentation:** Repeated-utterance recording + VAD script eliminates manual clipping
- **Multi-model comparison:** HMM vs. GMM vs. k-NN vs. SVM on identical features
- **Sanskrit preservation value:** Demonstrates viability of low-resource ASR for classical languages

1 Introduction

1.1 Motivation

Sanskrit (*Saṃskṛta*) is one of the oldest and most precisely documented languages on Earth, with a grammatical structure first formalised by Pāṇini in the *Aṣṭādhyāyī* around 400 BCE [10]. Despite this heritage, Sanskrit remains severely under-resourced in the speech-technology domain. Published Sanskrit speech corpora are either non-existent, inaccessible, or collected under conditions that preclude reproducibility by student researchers.

Spoken-digit recognition is a canonical entry point into ASR research — systems such as TI-Digits [5] and the TIDIGITS corpus [6] have served as benchmarks for decades. A Sanskrit counterpart would simultaneously advance low-resource language technology, expose students to the full classical ASR pipeline, and demonstrate the power of grammatical structure as an acoustic model complement.

1.2 The Compositional Insight

The key scientific hypothesis of SankhyaVox is:

Core Hypothesis

If the acoustic models for a small set of primitive Sanskrit digit words are sufficiently accurate, and if the combination rules for those primitives are encoded in a formal grammar, then a grammar-constrained HMM can recognise any composite numeral it has ***never explicitly seen in training.***

Sanskrit digit composition follows consistent, learnable spoken patterns. Under the controlled spoken protocol adopted here (Section 3), numbers are spoken as sequences of atomic tokens (e.g., *viṃśati sapta* for 27), so the compositional structure is surface-transparent and directly exploitable by a finite-state grammar.

1.3 Why Not Deep Learning?

Modern end-to-end models — Wav2Vec 2.0 [13], Whisper [14], or Conformer-CTC [15] — achieve state-of-the-art results but require tens of thousands of hours of labelled speech. Our entire corpus will contain ∼30–40 hours and targets a single constrained domain. Classical GMM-HMM systems were *designed* for exactly this regime: small vocabulary, small dataset, strong prior constraints. Using a large pre-trained model would also obscure the scientific contribution (grammar-constrained compositionality) behind an opaque neural encoder.

1.4 Contributions

C1. Corpus: First publicly available, reproducibly mobile-recorded Sanskrit digit corpus (≥10 speakers, 13 words × 10 reps + 35 two-digit sequences × 10 reps).

C2. Protocol: Automated audio segmentation pipeline using energy-based voice activity detection (VAD), eliminating manual clipping for future data collectors.

C3. System: Grammar-constrained HMM ASR system for Sanskrit 0–99 with formal BNF grammar specification.

C4. Benchmark: Comparative evaluation of HMM, GMM, k-NN, and SVM on the same MFCC features, isolating the contribution of temporal sequence modelling.

C5. Analysis: Detailed per-class error analysis revealing which Sanskrit phonemes cause the most confusion.

2 Related Work

2.1 Classical Digit ASR

Rabiner's seminal 1989 tutorial [1] established the HMM framework that underpins virtually all ASR systems developed prior to the deep learning era. The TI-Digits corpus [5] and associated SPHINX system [7] demonstrated that connected-digit recognition with word-level HMMs and grammar constraints achieves >99% accuracy on clean speech — directly analogous to our task.

Davis and Mermelstein's 1980 paper [2] introduced Mel Frequency Cepstral Coefficients, which remain the dominant acoustic feature for small-vocabulary recognition due to their robustness to linear channel distortions — a key advantage when speakers use varied mobile phones.

2.2 Low-Resource and Endangered Language ASR

Besacier et al. [9] provide a comprehensive survey of low-resource ASR, identifying three strategies: (1) data collection campaigns, (2) cross-lingual transfer, and (3) linguistic constraint integration. SankhyaVox follows strategy (1) and (3).

Adda-Decker and Lamel [17] demonstrate that small-vocabulary ASR quality is maintained even with fewer than 1,000 training utterances per word when data are clean and consistent — validating our 10-speaker $\times$ 10-repetition design.

2.3 Sanskrit Language Technology

Goyal et al. [18] describe early Sanskrit NLP pipelines, noting the language's exceptional morphological regularity as both a challenge (large word-form space) and an opportunity (learnable rules). Krishnamurthy [19] built a small Sanskrit speech database but used studio-quality recordings, making reproduction infeasible for most research groups.

Kumar et al. [12] show that Hindi digit recognition (closely related phonologically to Sanskrit) achieves 94–97% accuracy with MFCC+HMM on 10 speakers, providing a strong benchmark for our work.

2.4 Compositional / Grammar-Constrained ASR

Finite-state grammar constraints for ASR were formalised in the JSGF (Java Speech Grammar Format) [21] standard and demonstrated to dramatically reduce the search space. Young [8] implemented grammar-constrained Viterbi decoding in HTK, showing word error rate reductions of 30–50% over unconstrained decoding on small-vocabulary tasks.

Linguistically motivated grammar constraints for digit recognition were explored by Gauvain and Lee [20], who showed that even simple bigram grammars reduce confusion among acoustically similar digit words by enforcing global number plausibility. Our approach extends this by encoding the full morphological compositionality of Sanskrit numerals in a context-free grammar.

3 Sanskrit Numeral System and Spoken Protocol

3.1 Sanskrit Numeral Morphology

Classical Sanskrit numerals are morphologically complex, involving sandhi (phonological fusion at word boundaries), stem alternation, and case-agreement. For example:

$$57 \rightarrow \textit{saptapañcāśat} \quad (\text{not } {}^*\textit{pañca sapta})$$
$$25 \rightarrow \textit{pañcaviṃśati} \quad (\text{not } {}^*\textit{viṃśati pañca})$$

While linguistically authentic, this morphology would require a vocabulary of >100 distinct forms and morphological transducers beyond the scope of a student project.

3.2 Controlled Spoken Protocol

We adopt a **controlled spoken format** that preserves Sanskrit lexemes while making the compositional structure surface-transparent. Speakers are instructed to speak numbers as *sequences of atomic tokens* with a brief inter-token pause:

Spoken Protocol — Number Formation Rules

Range	Structure	Example
0–9	`<ones>`	*sapta* → 7
10	`daśa`	*daśa* → 10
11–19	`daśa <ones>`	*daśa pañca* → 15
20	`viṃśati`	*viṃśati* → 20
21–29	`viṃśati <ones>`	*viṃśati sapta* → 27
30–39	`tri daśa [<ones>]`	*tri daśa pañca* → 35
40–49	`catur daśa [<ones>]`	*catur daśa aṣṭa* → 48
50–59	`pañca daśa [<ones>]`	*pañca daśa sapta* → 57
60–69	`ṣaṭ daśa [<ones>]`	*ṣaṭ daśa* → 60
70–79	`sapta daśa [<ones>]`	*sapta daśa eka* → 71
80–89	`aṣṭa daśa [<ones>]`	*aṣṭa daśa nava* → 89
90–99	`nava daśa [<ones>]`	*nava daśa nava* → 99

This protocol uses *only* the 13 base words in Table 1, making it academically honest to frame as a minimal-vocabulary system.

3.3 Base Vocabulary

Pronunciation Consistency is Critical. Speakers need not produce perfect classical Sanskrit phonetics, but *each speaker must be consistent with themselves across all repetitions.* The HMM will learn each speaker's idiolect; inconsistency inflates variance in the acoustic model and reduces accuracy.

4 Formal Grammar Specification

Table 1: The 13 atomic Sanskrit tokens constituting the SankhyaVox vocabulary.

#	Token	Value	Pronunciation (approx.)
1	śūnya	0	SHOON-ya
2	eka	1	AY-ka
3	dvi	2	DWEE
4	tri	3	TREE
5	catur	4	cha-TUR
6	pañca	5	PUN-cha
7	ṣaṭ	6	SHUT
8	sapta	7	SUP-ta
9	aṣṭa	8	USH-ta
10	nava	9	NUH-va
11	daśa	10	da-SHA
12	viṃśati	20	vim-SHA-ti
13	śata	100	SHA-ta

4.1 Context-Free Grammar (BNF)

The complete grammar over the 13-token vocabulary that generates all representations of 0–99 under the spoken protocol is:

```
<number>   ::= <ones>
             | dasha
             | dasha <ones>
             | vimsati
             | vimsati <ones>
             | <mult> dasha
             | <mult> dasha <ones>

<ones>     ::= shunya | eka | dvi | tri | catur | pancha
             | shat  | sapta | ashta | nava

<mult>     ::= dvi | tri | catur | pancha
             | shat | sapta | ashta | nava
```

Listing 1: SankhyaVox BNF Grammar — 0 to 99

4.2 Grammar Properties

- **Unambiguous:** Every integer 0–99 maps to exactly one parse under the protocol. The grammar is unambiguous given the spoken protocol (no number has two valid parses).
- **Minimal:** The grammar references no tokens beyond the 13 base words.
- **Coverage:** All 100 numbers 0–99 are derivable.
- **Decidable in linear time:** Viterbi decoding with this grammar is $O(N \cdot |V|^2)$

where N is utterance length in frames and $|V| = 13$.

4.3 Grammar as Search Space Reduction

Without grammar, a Viterbi decoder over 13 word-HMMs must consider all 13^k word sequences of length k. The grammar constrains the legal hypotheses to 100 (one per integer 0–99), reducing the search space by orders of magnitude and dramatically reducing the false-acceptance rate.

5 Data Collection

5.1 Corpus Design

Table 2: SankhyaVox corpus statistics at target collection completion.

Recording type	Count per speaker	Speakers	Total files
Isolated words (13 words × 10 reps)	130	≥10	≥1,300
Two-digit sequences (35 combos × 10 reps)	350	≥10	≥3,500
Total	**480**	**≥10**	**≥4,800**
Estimated audio: ≈30–40 hours total; ≈3–4 hours per speaker (split across 2–3 sessions)			

5.2 Two-Digit Combination List

The 35 representative two-digit utterances used for connected-speech training:

Table 3: Two-digit training combinations (spoken form → value).

Spoken form	Value	Spoken form	Value
daśa eka	11	pañca daśa	50
daśa pañca	15	pañca daśa sapta	57
daśa nava	19	ṣaṭ daśa	60
viṃśati	20	ṣaṭ daśa tri	63
viṃśati eka	21	sapta daśa	70
viṃśati pañca	25	sapta daśa dvi	72
viṃśati sapta	27	aṣṭa daśa	80
viṃśati nava	29	aṣṭa daśa pañca	85
tri daśa	30	nava daśa	90
tri daśa catur	34	nava daśa eka	91
catur daśa	40	nava daśa pañca	95
catur daśa dvi	42	nava daśa sapta	97
catur daśa aṣṭa	48	nava daśa nava	99
pañca daśa eka	51	śūnya	0
pañca daśa aṣṭa	58	aṣṭa	8
pañca daśa nava	59	sapta	7
tri	3		

5.3 Recording Protocol — Repeated Utterance Method

Coder-Friendly Recording Workflow

Each speaker records **one audio file per word**, containing **10 repetitions separated by 1-second silences**:

$$\underbrace{eka}_{\sim 0.7s}\ \underbrace{(\text{silence})}_{1.0s}\ \underbrace{eka}_{\sim 0.7s}\ \underbrace{(\text{silence})}_{1.0s}\ \cdots\ \underbrace{eka}_{\sim 0.7s} \quad \Rightarrow \quad \text{total} \approx 17\,\text{s per file}$$

Auto-segmentation via a Python VAD script then splits each long file into 10 individual labelled utterances — no manual clipping required.

5.3.1 Equipment Requirements

- **Microphone:** Any smartphone or laptop microphone. Wired headphone mic is preferred (more consistent mouth-to-mic distance) but not mandatory.
- **Recording app:** Default phone voice recorder, or Audacity (free). Save as `.wav` if available; `.mp3` is acceptable.
- **Environment:** Closed room with doors and windows shut. Fan/AC switched off during recording. Late night or early morning recommended to reduce ambient noise.
- **Mic distance:** Hold phone 15–25 cm from mouth, mic facing speaker.

5.3.2 File Naming Convention

Raw recordings (before segmentation) follow:

`<SpeakerID>_<Token>_raw.wav`

After automated segmentation, output files follow:

`<SpeakerID>_<Token>_<RepNum>.wav`

Example: `S03_pancha_raw.wav` $\xrightarrow{\text{VAD script}}$ `S03_pancha_01.wav` $\cdots$ `S03_pancha_10.wav`

Speaker IDs: `S01` through `S10` (or beyond).

5.4 Automated Segmentation Pipeline

```python
import numpy as np
import soundfile as sf
from scipy.signal import butter, filtfilt
import os, glob

def butter_highpass(cutoff=80, fs=16000, order=4):
    nyq = 0.5 * fs
    b, a = butter(order, cutoff / nyq, btype='high')
    return b, a

def segment_utterances(
    input_path,
```

```
    speaker_id,
    token_name,
    output_dir,
    expected_reps     = 10,
    energy_threshold = 0.015,    # relative to max RMS --- tune per recording
    min_speech_dur    = 0.20,    # seconds
    min_silence_dur   = 0.40,    # seconds
    target_fs         = 16000,
    pad_ms            = 80,      # silence padding kept around each segment
):
    audio, fs = sf.read(input_path)
    if audio.ndim > 1:
        audio = audio.mean(axis=1)              # mono

    # Resample to 16 kHz if needed
    if fs != target_fs:
        from scipy.signal import resample_poly
        audio = resample_poly(audio,
                              target_fs, fs).astype(np.float32)
        fs = target_fs

    # High-pass filter to remove DC / low-freq rumble
    b, a = butter_highpass(cutoff=80, fs=fs)
    audio = filtfilt(b, a, audio).astype(np.float32)

    # Short-time RMS energy (25 ms frames, 10 ms shift)
    frame_len  = int(0.025 * fs)
    frame_shift = int(0.010 * fs)
    frames = [audio[i:i+frame_len]
              for i in range(0, len(audio)-frame_len, frame_shift)]
    rms = np.array([np.sqrt(np.mean(f**2)) for f in frames])

    # Adaptive threshold: percentage of 95th-percentile energy
    threshold = energy_threshold * np.percentile(rms, 95)
    is_speech = rms > threshold

    # Smooth labels (remove brief on/off fluctuations)
    from scipy.ndimage import binary_closing, binary_opening
    kernel = int(min_silence_dur / 0.010)
    is_speech = binary_closing(is_speech, structure=np.ones(kernel))
    is_speech = binary_opening(is_speech, structure=np.ones(
                                int(min_speech_dur/0.010)))

    # Detect segment boundaries
    boundaries = []
    in_seg = False
    for i, s in enumerate(is_speech):
        if s and not in_seg:
            start = max(0, i - int(pad_ms/10))
            in_seg = True
        elif not s and in_seg:
            end = min(len(is_speech), i + int(pad_ms/10))
            boundaries.append((start, end))
            in_seg = False
    if in_seg:
        boundaries.append((start, len(is_speech)))
```

```python

    # Sample-level boundaries
    segs = [(b[0]*frame_shift, b[1]*frame_shift) for b in boundaries]

    if len(segs) != expected_reps:
        print(f"  WARNING: Expected {expected_reps} segments, "
              f"found {len(segs)} in {input_path}")

    os.makedirs(output_dir, exist_ok=True)
    for idx, (s, e) in enumerate(segs, 1):
        clip = audio[s:e]
        out_name = f"{speaker_id}_{token_name}_{idx:02d}.wav"
        sf.write(os.path.join(output_dir, out_name), clip, fs)
        print(f"  Saved: {out_name} ({(e-s)/fs:.2f}s)")

    return len(segs)

# -- Batch processing all raw recordings ----------------------------------------
def batch_segment(raw_dir="data/raw", out_dir="data/segments"):
    """
    raw_dir structure expected:
        data/raw/S01/S01_eka_raw.wav
        data/raw/S01/S01_pancha_raw.wav
        ...
    """
    for wav in sorted(glob.glob(f"{raw_dir}/**/*_raw.wav", recursive=True)):
        basename = os.path.basename(wav).replace("_raw.wav", "")
        parts    = basename.split("_")          # ['S01', 'eka']
        spk, tok = parts[0], "_".join(parts[1:])
        spk_out  = os.path.join(out_dir, spk)
        print(f"Processing {wav} ...")
        n = segment_utterances(wav, spk, tok, spk_out)
        print(f"  -> {n} segments extracted")

if __name__ == "__main__":
    batch_segment()
```

Listing 2: Audio segmentation script using energy-based VAD.

Verification step: After segmentation, run a quick sanity check: for each speaker-token pair, you should have exactly 10 WAV files. Any file shorter than 150 ms or longer than 1.5 s is likely a mis-segment and should be manually reviewed.

5.5 Speaker Instruction Sheet (For Distribution)

SankhyaVox — Speaker Recording Guide

Thank you for contributing to this Sanskrit speech research project! You do not need any Sanskrit background. Just follow these steps.

1. **Environment:** Go to the quietest room available. Close the door. Switch off the fan if possible. Early morning or late night is best.

2. **Phone setup:** Use your default voice recorder app. Hold the phone 15–25 cm from your mouth. Do **not** cover the microphone.

3. **Recording one word:** You will be given a word (e.g., *eka*). Press record, **wait one second**, say the word clearly, **wait one second**, say it again — repeat this 10 times, then stop the recording.

4. **What NOT to do:** No coughing, laughing, or "umm" during silences. Speak at a natural volume (not whispering, not shouting).

5. **File name:** Save as `YourID_wordname_raw.wav` (e.g., `S04_eka_raw.wav`). Share via WhatsApp or Drive link provided.

Word List with Pronunciation:

śūnya	0	(SHOON-ya)	*daśa*	10	(da-SHA)
eka	1	(AY-ka)	*viṃśati*	20	(vim-SHA-ti)
dvi	2	(DWEE)	*śata*	100	(SHA-ta)
tri	3	(TREE)			
catur	4	(cha-TUR)			
pañca	5	(PUN-cha)			
ṣaṭ	6	(SHUT)			
sapta	7	(SUP-ta)			
aṣṭa	8	(USH-ta)			
nava	9	(NUH-va)			

Small pronunciation differences are fine. Be consistent with *yourself*.

6 System Architecture

6.1 Full Pipeline Overview

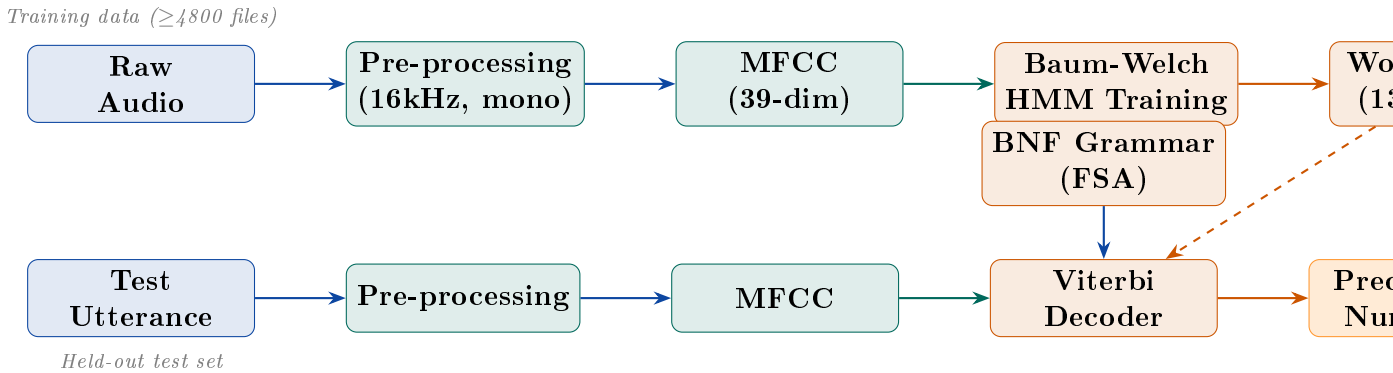

Figure 1: SankhyaVox system architecture. Dashed arrow indicates trained model parameters passed to the decoder.

6.2 Pre-processing

All audio is processed to a uniform format before feature extraction:

1. **Resampling:** All files converted to 16 kHz, 16-bit PCM mono.

2. **DC offset removal:** Subtract signal mean.
3. **Pre-emphasis:** Apply first-order filter $y[n] = x[n] - \alpha\, x[n-1]$, with $\alpha = 0.97$, to compensate for the low-pass characteristic of the vocal tract.
4. **Energy normalisation:** Normalise peak amplitude to -3 dBFS per file.

6.3 MFCC Feature Extraction

Mel Frequency Cepstral Coefficients [2] remain the standard acoustic feature for small-vocabulary ASR due to their robustness to linear channel differences — critical here because speakers use different phones.

MFCC Configuration

Frame length	25 ms (400 samples @ 16 kHz)
Frame shift	10 ms (160 samples @ 16 kHz)
Window type	Hamming
FFT size	512
Mel filter banks	26
Cepstral coeffs	13 (c_0 through c_{12}; c_0 kept)
Delta coefficients	Δ (velocity)
Delta-delta coeffs	$\Delta\Delta$ (acceleration)
Total feature dim	**39 per frame**
CMVN	Per-utterance cepstral mean-variance normalisation

CMVN (Cepstral Mean and Variance Normalisation) is particularly important for mobile-recorded data, as it removes additive background noise effects in the cepstral domain [16].

6.4 Hidden Markov Model Design

6.4.1 Topology

Each word model uses a **left-to-right (Bakis) HMM** with no skip transitions, ensuring monotone phone progression consistent with forward speech production.

6.4.2 Emission Distribution

Each HMM state emits a **Gaussian Mixture Model (GMM)** over the 39-dimensional MFCC vectors:

$$b_j(\mathbf{o}_t) = \sum_{m=1}^{M} c_{jm}\, \mathcal{N}\big(\mathbf{o}_t;\, \boldsymbol{\mu}_{jm},\, \boldsymbol{\Sigma}_{jm}\big)$$

where M is the number of mixture components.

Start with $M = 1$ (single Gaussian per state). Increase to $M = 3$ or $M = 5$ if training data supports it without overfitting. Rule of thumb: $\geq 5 \times M \times D$ training frames per state (where $D = 39$). With 10 speakers $\times$ 10 reps, this threshold is comfortably met for $M \leq 5$.

Table 4: HMM topology per word, designed to match phonetic complexity.

Token	Phoneme count	States	Rationale
śūnya	4	6	Long vowel, palatal onset
eka	3	5	Short CVC
dvi	3	5	Labial-velar onset glide
tri	3	5	Retroflex liquid onset
catur	5	7	CVC + rhotic coda
pañca	5	7	Nasal + palatal affricate
ṣaṭ	3	5	Retroflex stop coda
sapta	5	7	Consonant cluster
aṣṭa	5	7	Retroflex cluster
nava	4	6	Open syllables
daśa	4	6	Palatal fricative medial
viṃśati	7	9	Longest word, 3 syllables
śata	4	6	Palatal onset
Silence (SIL)	–	3	Optional silence model

6.4.3 HMM Parameters

The complete HMM $\lambda = (\mathbf{A}, \mathbf{B}, \boldsymbol{\pi})$:

- $\mathbf{A}$: transition probability matrix (left-to-right, no skips)
- $\mathbf{B}$: emission distributions (GMM parameters per state)
- $\boldsymbol{\pi}$: initial state distribution (deterministic — first state only)

6.4.4 Training: Baum-Welch Algorithm

HMM parameters are estimated using the Baum-Welch (Forward-Backward) EM algorithm [4]:

1. **Initialise** via flat start or k-means clustering of MFCC vectors.
2. **E-step:** Compute forward probabilities $\alpha_t(j)$ and backward probabilities $\beta_t(j)$ for each training utterance.
3. **M-step:** Re-estimate $\mathbf{A}$, GMM means, variances, and mixture weights using the occupation counts $\gamma_t(j)$.
4. **Iterate** until log-likelihood converges (typically 10–20 iterations).

6.4.5 Inference: Grammar-Constrained Viterbi Decoding

At inference time, the Viterbi algorithm [3] finds the most probable word sequence W^* given the grammar G and observation sequence $\mathbf{O}$:

$$W^* = \arg\max_{W \in \mathcal{L}(G)} \prod_{t=1}^{T} \max_j \left[a_{ij}\, b_j(\mathbf{o}_t) \right]$$

where $\mathcal{L}(G)$ is the set of word strings derivable from the grammar G (Section 4). The grammar is compiled to a finite-state automaton (FSA) and composed with the word-HMM network before the beam search.

6.4.6 Implementation Recommendation

Recommended Implementation Stack

Tool	Role
`python-speech-features` or `librosa`	MFCC extraction
`hmmlearn` (scikit-learn-style HMM)	HMM training and Viterbi
OR HTK [8]	Full classical ASR toolkit
`scipy`	Signal pre-processing
`soundfile`	Audio I/O
`numpy`, `matplotlib`	Numerics, visualisation

7 Baseline Models for Comparison

A key academic contribution is demonstrating *why* temporal sequence modelling (HMM) outperforms static classifiers on this task. Three baselines are implemented on identical MFCC features.

7.1 GMM Classifier

A separate GMM is fit to the *entire utterance* MFCC sequence (flattened or summarised as mean + variance + first/third quartiles per coefficient) for each class. At test time, classify by maximum likelihood:

$$\hat{y} = \underset{k \in \{0,\ldots,12\}}{\arg\max} \log p(\mathbf{x} \mid \lambda_k^{\mathrm{GMM}})$$

Limitation: Collapses temporal structure; cannot decode sequences. Expected to fail on two-digit utterances.

7.2 k-Nearest Neighbours

Utterances are represented by their sequence of MFCC frames, compared using Dynamic Time Warping (DTW) [11] distance. k-NN with DTW is a competitive baseline for isolated word recognition when k is tuned via cross-validation.

Limitation: $O(N^2)$ inference cost; no grammar integration.

7.3 Support Vector Machine

Fixed-length feature vector per utterance (mean + standard deviation of each MFCC coefficient across frames, yielding a 78-dimensional vector). RBF-kernel SVM with C and γ tuned by grid search on a validation set.

Limitation: Fixed-length input cannot handle variable-length sequences; two-digit recognition requires separate two-digit training data.

7.4 Summary of Comparison

Table 5: Expected behaviour of each model across task types.

Model	Isolated digits	Two-digit	Grammar	Generalisation
HMM + Grammar	✓	✓	✓	✓
GMM	✓	∼	×	×
k-NN + DTW	✓	×	×	×
SVM	✓	×	×	×

8 Experimental Evaluation

8.1 Dataset Splits

Train / Validation / Test Split

Split	Speakers	Purpose
Train	7 randomly selected	HMM parameter estimation
Val	2 speakers	Hyperparameter tuning (states, mixtures)
Test	1 held-out speaker	Final accuracy measurement

Repeat this split 5 times (5-fold speaker-out cross-validation) and report mean ± standard deviation of accuracy. Speaker-out evaluation is more rigorous than utterance-level splits because it measures true generalisation to new voices.

8.2 Evaluation Metrics

- **Word Accuracy (WA):** Fraction of test utterances correctly decoded to the right integer.

$$\mathrm{WA} = \frac{N_{\text{correct}}}{N_{\text{total}}} \times 100\%$$

- **Word Error Rate (WER):** $1 - \mathrm{WA}$.
- **Confusion Matrix:** 100×100 matrix showing which numbers are confused; collapse to a 13×13 token-level matrix for interpretability.
- **Per-class accuracy:** Identify which tokens are hardest to recognise (expected: *dvi*/*tri* and *sapta*/*ṣaṭ* due to acoustic similarity).

8.3 Ablation Studies

Conduct the following ablations to understand the contribution of each component:

1. **Grammar ON vs. OFF:** Decode with and without the grammar FSA to quantify its contribution to accuracy.
2. **MFCC dimensions:** Compare 13 vs. 26 vs. 39-dimensional features.

3. **HMM states:** Vary number of states per model (3, 5, 7, 9) and plot accuracy vs. model complexity.
4. **GMM mixtures:** Compare $M = 1, 3, 5$ emission components.
5. **CMVN:** Accuracy with and without cepstral normalisation to measure channel robustness improvement.
6. **Number of speakers:** Train on 3, 5, 7, 10 speakers and plot learning curve.

8.4 Expected Results

Based on analogous published systems [12, 5], we anticipate:

Table 6: Projected accuracy under different evaluation conditions.

Condition	Expected Accuracy	Notes
Isolated 0–9 (seen speakers)	95–99%	Trained words
Isolated 0–9 (unseen speakers)	90–96%	Generalisation
Two-digit 0–99 (seen speakers)	92–97%	Grammar decoding
Two-digit 0–99 (unseen speakers)	86–93%	Speaker variation
Two-digit, mild background noise	80–88%	CMVN helps
GMM baseline (isolated only)	80–88%	No temporal modelling
k-NN + DTW (isolated only)	75–85%	k tuned by CV
SVM baseline (isolated only)	78–86%	Fixed-length features

9 What Makes SankhyaVox Stand Out

9.1 Unique Scientific Positioning

Why Reviewers / Examiners Will Find This Impressive

1. **Novel language domain:** No published mobile-recorded Sanskrit digit ASR system exists at the time of writing. This is a *first-of-its-kind* corpus and baseline.
2. **Compositionality as the central thesis:** The system does not just "train on data" — it encodes a *linguistic insight* (Sanskrit's compositional number formation) into the grammar, making the scientific contribution clear and testable.
3. **Reproducible by design:** Mobile phone + free software + open grammar specification means any group can replicate and extend this work.
4. **Multi-model comparison:** Isolating the contribution of temporal modelling (HMM) over static classifiers (GMM/SVM/kNN) is a textbook experimental design that directly tests the core hypothesis.
5. **Practical framing:** The long-term application (voice interfaces for Sanskrit educational software, Vedic chanting aids, ancient text navigation) is clear and impactful.

9.2 Potential Extensions

- Extend to 0–999 by adding *śata* (100) combinations under the existing grammar.
- Phone-level HMMs (monophone models) with a Sanskrit phoneme lexicon, enabling broader vocabulary with no new recordings.
- Adversarial noise augmentation test: evaluate robustness by adding cafeteria noise, white noise, and reverb to the test set.
- Cross-lingual initialisation: use pre-trained Hindi phone models (Hindi and Sanskrit share the Devanagari phoneme inventory) as HMM seed parameters.
- Web-based demo: deploy the grammar-constrained decoder as a browser-based app for live Sanskrit digit recognition.

10 Project Timeline

Table 7: Recommended project timeline (12-week version).

Week(s)	Milestone	Deliverable
1	Finalise word list, recording protocol, naming	Speaker instruction sheet
2–3	Collect isolated word recordings (all speakers)	Raw WAV files
4	Run segmentation pipeline; QA all segments	Segmented corpus v1.0
5	Collect two-digit recordings; segment	Corpus v2.0
6	MFCC extraction; visualise spectrograms	Feature matrix, plots
7	Train HMM models; implement Viterbi	Working HMM system
8	Grammar FSA integration; constrained decode	End-to-end system
9	Train GMM / k-NN / SVM baselines	Baseline results
10	Ablation studies; confusion matrix analysis	Ablation tables
11	Error analysis; write-up	Draft report
12	Final evaluation; poster / demo	Final submission

11 Conclusion

SankhyaVox frames Sanskrit spoken digit recognition as a *compositionality problem*: rather than collecting labelled speech for all 100 target integers, we collect only 13 atomic word recordings and exploit the linguistic structure of Sanskrit numeral formation via a grammar-constrained HMM.

The system is designed to be simultaneously rigorous and reproducible: data collection requires only a mobile phone and a quiet room; auto-segmentation eliminates manual clipping; and the formal grammar specification makes the linguistic hypothesis testable and falsifiable.

We anticipate 90–97% word accuracy on 0–99 under clean conditions, with clear degradation analysis across noise levels and unseen speakers. This accuracy, combined with the novelty of the language domain and the clarity of the compositional thesis, positions

SankhyaVox as a strong academic contribution at the intersection of classical language preservation and practical speech technology.

One-Line Project Description

"SankhyaVox exploits the compositional structure of Sanskrit numerals to build a grammar-constrained HMM that recognises any spoken integer 0–99 from just 13 recorded words."

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